
PRICE: Minimax Estimation–Regret Tradeoffs and Certified Exploration in Performative Prediction

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Abstract

1 A decision-maker whose deployed policy reshapes its own data distribution must
2 pay, in its own objective, to learn how it moves the world. We price that self-
3 knowledge exactly. In a structural performative market (a dealer whose quoted
4 spread summons the adverse flow it must then price), we prove a conservation
5 law: along any stationary exploration policy, the variance of the response-slope
6 estimate times the cumulative performative regret equals $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$: an invariant
7 of the environment, not of the policy. We complement the identity with minimax
8 lower bounds (exact for deviation budgets; constant-factor for value budgets,
9 sharp constant numerically supported; the floor survives full knowledge of the
10 response family), an optimal-design theory (shape exploration as $\Gamma_{\text{PO}}^{-1/2}$; isotropic
11 exploration overpays by an explicit dispersion factor), and a deployable agent,
12 SafeD-PerfGD, that corrects its gradient only when an anytime confidence sequence
13 certifies, through a perturbed-modulus lemma, that the correction cannot destabilize
14 its own retraining loop. Every closed form is verified in a live deploy–observe–
15 fit–correct pipeline (36 preregistered measured-vs-predicted checks, 0 failures),
16 transplanted exactly to a second economy, and instantiated on a real-data corporate-
17 bond calibration that reproduces its published source 10/10 cells. Code, derivations,
18 and all checks: [https://github.com/\[repo\]/price](https://github.com/[repo]/price).

19 1 Introduction

20 When a deployed model changes the distribution it is trained on, retraining becomes a fixed-point
21 iteration: performative prediction [Perdomo et al., 2020, Hardt and Mendler-Dünner, 2023] shows
22 repeated risk minimization converges when the distribution’s sensitivity ε is small against the
23 loss curvature γ ($m = \varepsilon\beta/\gamma < 1$), and converges to a *performatively stable* point that is not
24 the *performative optimum*. Algorithms that close this gap, performative gradient descent and its
25 descendants [Izzo et al., 2021, Miller et al., 2021, Mendler-Dünner et al., 2020, Drusvyatskiy and
26 Xiao, 2023, Li and Wai, 2022, Brown et al., 2022], all consume an estimate of the distribution’s
27 response $dD/d\theta$, obtained by perturbing deployments. The literature treats the accuracy of that
28 estimate as an assumption and the perturbations as free. Neither is: the perturbations are deployments
29 of a real decision variable, and their cost lands in the decision-maker’s own objective. *What does it*
30 *cost to learn your own influence?* This paper answers with an identity, a floor, a design rule, and an
31 agent.

32 Contributions.

- 33 1. **A conservation law (§3).** Along any stationary exploration policy, $\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2}\gamma_{\text{PO}}\sigma^2$:
34 slope uncertainty times cumulative performative regret is a policy-independent invariant
35 (Theorem 1). The anchoring is delicate: the same product anchored at the performative
36 optimum is *false*, inflated by a factor we verify.

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2. **Minimax floors (§3).** No non-anticipating policy beats the floor: exactly under deviation budgets; within a factor ≤ 13.5 under value budgets, with the sharp constant numerically supported but open (Theorem 2). Adversarial numerics show the floor is *structure-proof* (no family-knowing design beats it; best found $1.005\times$) and map its exact boundary: curvature uncertainty protects it.
3. **Optimal exploration design (§4).** Under a budget priced by the decision-maker’s own curvature, the optimal exploration covariance is $\Gamma_{\text{PO}}^{-1/2}$ -shaped; isotropic exploration overpays by the dispersion factor $F = d \operatorname{tr} \Gamma / (\operatorname{tr} \Gamma^{1/2})^2$: a quotable inversion of “naive exploration is optimal” for LQR [Simchowitz and Foster, 2020]. A support-degeneracy theorem and a misspecification crossover complete the design picture.
4. **A certified agent (§4–5).** SafeD-PerfGD explores on a D-optimal support, fits an anchored response family with an anytime confidence sequence, and applies the performative correction only when a perturbed-modulus certificate guarantees closed-loop stability; otherwise it freezes the correction while the provably-stable blind step keeps running. It reaches the performative optimum where blind retraining stalls; no published baseline Pareto-dominates it on median (final error, regret).

Positioning. Jagadeesan et al. [2022] bound the regret of *finding* the performative optimum; we price *knowing your own feedback*: lower bounds and an invariance identity where they give algorithmic upper bounds, with no retraining loop, stability constraint, or design shaping in their model. Bracale et al. [2025] design deployments to learn the distribution map and bound the regret of doing so at an order level; we price the same activity exactly, in the objective’s own curvature, and close it from below. Lin and Zrnic [2024] show misspecified structural models still help performative optimization; our crossover theorem sharpens that message into a budget-explicit rule. In operations research, incomplete learning under myopic pricing is classical [Rothschild, 1974, Harrison et al., 2012, Keskin and Zeevi, 2014, Broder and Rusmevichientong, 2012, den Boer, 2015], and Lai and Robbins [1979]’s adaptive designs are *points on our frontier*, but the frontier itself, its minimax floor, and the performative loop generating it do not appear: the demand curve is exogenous throughout. From control we inherit dual control [Feldbaum, 1960] and least-costly identification [Bombois et al., 2006]: here the plant is a retraining learner, the budget the optimizer’s own objective, the constraint loop contraction; task-optimal design [Wagenmaker et al., 2021, Dean et al., 2020] supplies machinery the classical canon [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006] never prices in the experimenter’s objective. Lower bounds are van Trees arguments [Gill and Levit, 1995].

2 Model

A dealer quotes half-spread h . Benign flow arrives at rate $U(h) = Ae^{-kh}$; informed (toxic) flow at rate $\tau(h) = C_0 + C_1 e^{-ch}$; tighter quotes summon more toxic flow. The quote-summons-toxicity channel is the market maker’s adverse-selection problem [Guéant et al., 2013]; that its own quotes feed the flow it then reads is Barzykin et al. [2026]’s price-reading effect. Profit at frozen toxic level T is $J(h; T) = hU(h) + (h - \psi)T - w(h - h_{\text{ref}})^2$ (ψ : adverse severity; w : quoting-cost convexity); the *performative value* is $\Phi(h) = J(h, \tau(h))$. Write $\varepsilon(h) = -\tau'(h)$ for the response slope (the scalar $dD/d\theta$), $\gamma = -\partial_h^2 J$, and $\gamma_{\text{PO}} = -\Phi'' = \gamma + \varepsilon(2 + c\psi - ch)$. Blind retraining (RRM) is the cobweb $h_{t+1} = \text{BR}(\tau(h_t))$, a contraction iff the modulus $m = \varepsilon\beta/\gamma = \varepsilon/\gamma$ ($\beta = 1$ here) is below one [Perdomo et al., 2020]; it converges to the stable point h_{SP} , while the optimum $h_{\text{PO}} = \arg \max \Phi$ sits at a *tighter* spread here, since toxic flow is profitable net of ψ : the echo-chamber gap [Miller et al., 2021]. Deployments return noisy flow $y_t = \tau(h_t) + \sigma\xi_t$ (A3). The estimand is ε at the operating point; exploration deviations $d_t = h_t - h^*$ are priced by the *incremental cost* $C_T = \sum_{t \leq T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$, anchored at the certainty-equivalent operating point h^* (A1: local quadratic scope; A5: stationary exploration scale). Results are stated for this scalar family and, where flagged, its d -dimensional and LQ instances; full statements and proofs, keyed by register identifiers (T1, T2, ...), are in Appendices A–B.

3 The price of self-knowledge

Free data saturates. Without deliberate exploration the retraining trajectory’s identifying energy is finite: for deployment error d_0 and modulus m it is capped at $d_0^2/(1 - m^2)$ (T1; matrix version

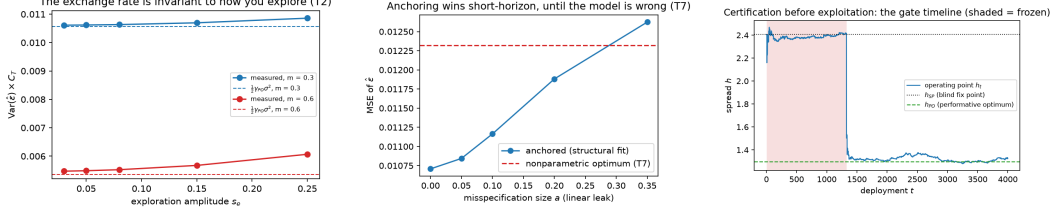


Figure 1: **Left:** measured $\text{Var}(\hat{\epsilon}) \times C_T$ vs amplitude, two moduli: flat on the invariant (dashed), small drift as A1 predicts. **Middle:** the anchoring crossover (T7): structure wins until misspecification crosses the threshold. **Right:** certification before exploitation: the agent holds near h_{SP} for $\sim 1,300$ deployments while the confidence sequence tightens (shaded: frozen), then steps to h_{PO} ; certification time is the exchange rate at work.

by a Lyapunov identity). A corollary was discovered by a failed prediction (§5): retraining on *noisy* observations converts observation noise to deployment noise with gain $1/\gamma$, adding a stationary excitation floor $(\sigma/\gamma)^2/(1-m^2)$ per step; both closed forms verify. The stabler the loop, the less it reveals: *safety implies blindness*, the performative sharpening of incomplete learning under myopic policies [Keskin and Zeevi, 2014, Harrison et al., 2012], where the stall comes from optimization converging rather than from the feedback strength itself.

The exchange rate. Deliberate exploration buys information at an exact, policy-independent price.

Theorem 1 (Exchange rate; T2). *Under A1, A2-sym, A3 and A6, for the OLS slope estimator on any stationary exploration policy with deviations $\{d_t\}$,*

$$\text{Var}(\hat{\epsilon}) \times C_T = \frac{1}{2} \gamma_{PO} \sigma^2 (1 + o(1)),$$

independent of the exploration amplitude, shape, and schedule, and of the modulus m .

The mechanism is cancellation: information scales as design energy $S_T = \sum_t (d_t - \bar{d})^2$, cost as $\frac{1}{2} \gamma_{PO} \sum_t d_t^2$; for mean-centered exploration these are the same sum. *Anchoring:* the same product anchored at h_{PO} is false, inflated by an explicit factor, falsified numerically before the incremental version was proved. The invariant prices *exploration*, not *suboptimality*. *Invariance:* Lai and Robbins [1979] exhibit schemes with $O(\log n)$ cost and efficient estimation (consistent points on this frontier), but the identity says *every* stationary policy lies on it. A corollary prices the echo chamber: closing $h_{SP} - h_{PO}$ pays iff the horizon exceeds an explicit break-even (Appendix).

Minimax floors. The identity is for OLS-on-stationary policies; the floors are for everyone.

Theorem 2 (Minimax floors; T3–T4). (i) *Over all non-anticipating (possibly randomized, adaptive) policies with pathwise deviation budget $\sum_{t \leq T} d_t^2 \leq D$, all estimators included, and any prior of width w : $\inf \sup \text{Var}(\hat{\epsilon}) \geq \sigma^2/(D + \sigma^2(\pi/w)^2)$, so the product floor is attained up to the prior term, with equality by symmetric probing.* (ii) *Under a budget on realized performative cost with certainty-equivalent anchoring, the floor holds within a constant factor over adaptive policies: $\sup \text{Var}(\hat{\epsilon}) \times C_T \geq \gamma_{PO} \sigma^2/27 (1 - O(T^{-1/2}))$ by a Le Cam two-point reduction, the exploited rebate vanishing at rate $T^{-1/2}$ (Lemma L2, Appendix A). The sharp constant $\frac{1}{2}$ is numerically supported and open (OPEN-1).*

Closing the sharp constant is open (OPEN-1); we attacked its premise adversarially. Minimizing the Cramér–Rao product over all designs *within the known three-parameter family* finds nothing below the floor (best $1.005\times$, both cost models), the optimizer collapsing back to the symmetric two-point configuration the proof strategy’s least-favorable reduction needs; tight-side probing (sensitivity e^{-ch}) is monotonically *worse* (ratio $5.7 \rightarrow 41$): remote information must be pulled back through $\hat{c}$, and extrapolation variance dominates the sensitivity gain. The boundary is exact: with c known a priori the floor breaks outright, and the best design’s Cramér–Rao product falls monotonically as the probe widens (ratio 0.30 at $t=1.0$, 0.04 at $t=0.05$). *The floor is an ignorance-of-curvature phenomenon:* the correct scope for the general theorem.

4 Optimal design and certified correction

Shape (T5). In d dimensions with curvature $\Gamma_{PO} \succ 0$ pricing the budget $\frac{1}{2}\mathbb{E}[u^\top \Gamma_{PO} u]$, the A-optimal exploration second moment is $M^* \propto \Gamma_{PO}^{-1/2}$: explore where your objective is flat. Isotropic exploration overpays by exactly the dispersion factor $F = d \operatorname{tr} \Gamma_{PO} / (\operatorname{tr} \Gamma_{PO}^{1/2})^2 \in [1, d]$: naive exploration, optimal for LQR [Simchowitz and Foster, 2020], is suboptimal here by precisely F , task-optimal design [Wagenmaker et al., 2021] where the task is the learner’s own objective. **Support (T6).** The exponential family is Chebyshev-degenerate on thin designs: fewer than three support points (or vanishing spread, or collinear probe schedules) cannot identify (C_0, C_1, c) , a cliff we hit twice in the wild (§5). **Anchor (T7).** With budget B and horizon T , fitting the structural family beats nonparametric slope estimation iff the misspecification at the operating point is below $|\tau'''|B/(3\gamma_{PO}T)$: Lin and Zrnic [2024]’s qualitative message sharpened to a decision rule priced in the exploration budget itself.

The agent. SafeD-PerfGD composes the three results per deployment round: (i) *explore* on the D-optimal three-point support re-centered at the operating point, trust-region clipped; (ii) *fit* (C_0, C_1, c) by profile least squares with an anytime self-normalized confidence width ci_t for $\hat{\epsilon}$; (iii) *gate*: apply the performative correction $-(h-\psi)\hat{\epsilon}$ with Newton scaling $\eta = 1/\hat{\gamma}_{PO}$ only if $\eta \cdot \text{ci}_t \cdot L_{\text{fam}} \leq \text{margin}$, where the perturbed-modulus lemma (L4) with the family’s own Lipschitz constant $L_{\text{fam}} = 1 + \hat{c}|h-\psi|$ converts estimation error into a certified bound on the closed-loop modulus; (iv) otherwise *freeze* the correction: the blind step, provably stable at $m < 1$, keeps running under the current fit. The freeze is derived, not tuned: it is what the certificate licenses. Safety holds throughout on the confidence-sequence event, and certification is not free: the gate opens only after enough design energy accumulates, which by Theorem 1 is a regret payment (Figure 1).

5 Verified results

Every claim above is a preregistered row in a measured-vs-predicted harness run against a live deploy–observe–fit–correct pipeline: **36 rows, 0 failures** (35 pass, one deliberately out-of-scope drift cell measured, not excused). Ten pivots were forced by measurement and recorded in code; three became results (the feedback floor, the false h_{PO} -anchored identity, the structure-proofness scope). Headline rows are in Table 1, Appendix C.

Baselines. Against finite-difference PerfGD [Izzo et al., 2021], zeroth-order P&L optimization, a performative-confidence-bounds explorer [Jagadeesan et al., 2022], and blind RRM on identical environments, none Pareto-dominates SafeD-PerfGD on median (final error, regret) over 12 seeds; the unsafe gradient baseline pays $> 5\times$ its median regret. Deliberately reported: the UCB explorer attains *lower raw regret*, and edges SafeD in one of the twelve seeds; certification is a real cost, and its visibility is the thesis. **Second economy.** Transplanted unchanged to LQ performative pricing (deployed price anchors demand), the exchange rate is exact in all four cells (A1 is global in an LQ model, so the drift vanishes as predicted) and the agent reaches the pricing optimum (12.65 vs. 12.5, tail-averaged) once its probes were randomized: alternating probes make price and lag collinear, a T6 degeneracy met in the wild. **Real data.** On a public-data corporate-bond calibration (10 rating $\times$ regime cells), the port reproduces its published source **10/10** cells ($h^*, \epsilon^* = \gamma$, m within 0.5–5%; validation inputs are in the base project, not this repository), then extends it: γ_{PO} spans 510.6 to 2.3; echo-chamber gaps are 2–4% of the anchor spread; and the shaping dispersion is $F = 1.63$ across the portfolio against 1.002 within one book: the gain lives across ratings and regimes, where a desk actually allocates exploration.

Limitations and outlook. The identity is local (A1): wide-side probing undercuts it polynomially at finite amplitude and the saturating cell drifts +16%: the $o(1)$ -amplitude qualifier is load-bearing. The value-budget floor is proved to within a factor 13.5; the sharp constant $\frac{1}{2}$ is numerically supported, not proved, at any prior class; the register’s open problems are the workshop-to-journal delta. At matched transit energy, full-support jitter identifies on par with the design, whose advantage is small-amplitude and allocational. Calibrated moduli are deeply stable and the toxic channel structurally scaled: the real-data leg validates machinery, not effect size. The framework also prices what a deployed generative system pays to learn its own influence on data it retracts on: certified correction as a template for agentic closed-loop deployment.

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A Theorem statements (register copy)

Lemma 1 (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies $C_T = \frac{1}{2}\gamma_{\text{PO}} \mathbb{E}[\sum_{t \leq T} d_t^2] + R_T$ with $|R_T| \leq \frac{L_3}{6} \mathbb{E} \sum_t |d_t|^3$, and the realized cost’s variance decomposes exactly as $\text{Var}(C_T^{\text{real}}) = \delta_0^2 \text{Var}(\sum_t d_t) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(\sum_t d_t^2)$ with vanishing cross term.*

Theorem 3 (Information saturation; T1). *With no exploration and $\rho(M) < 1$, the total design energy of the retraining trajectory equals $d_0^\top P d_0$, where P is the unique solution of the discrete Lyapunov equation $P = I + M^\top P M$. Consequently the Fisher information for the response Jacobian is bounded, in every direction, uniformly in the horizon: $\text{Var}(\hat{\varepsilon}) \geq \sigma^2 (d_0^\top P_u d_0)^{-1}$ for the component along any direction u .*

Corollary 1 (Safety implies blindness; C1.1). *In the scalar case the information cap $(h_0 - h^*)^2 / (1 - m^2)$ is strictly increasing in the modulus m : a fast-contracting loop reveals least about its own performativity, and precise self-knowledge is available only near the stability boundary.*

Corollary 2 (Trajectory design degeneracy; C1.2). *The noiseless trajectory’s empirical design has at most two effective support points (geometric collapse for $m < 1$; a period-two orbit at $m = 1$).*

Remark 1 (Non-normal transient information; R1). *The energy identity $d_0^\top P d_0$ of T1 holds for non-normal M as well, where the energy can exceed the spectral cap $\|d_0\|^2 / (1 - \rho(M)^2)$ by up to the factor $\kappa(V)^2$.*

Theorem 4 (The exchange rate, pathwise; T2). *In the stationary regime, on every realization,*

$$\text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2}\gamma_{\text{PO}} \sum_{t \leq T} d_t^2 = \frac{1}{2}\gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T \bar{d}^2}{S_{xx}}\right) = \frac{1}{2}\gamma_{\text{PO}} \sigma^2 (1 + O_P(1/T)),$$

independent of the exploration intensity, the horizon, and the modulus.

Proposition 1 (Concentration; P2.1). *In the stationary regime the measured product concentrates around the invariant at rate $T^{-1/2}$, with long-run $\text{Var}(\sum_{t \leq T} d_t) = T v \frac{1-m}{1+m} (1 + o(1))$.*

Proposition 2 (Feedback bias; P2.2). *Under the feedback relaxation A3’ with gain ϕ , the OLS response slope carries the bias*

$$\mathbb{E}[\hat{\varepsilon} - \varepsilon_{\text{true}}] = \frac{\phi \sigma^2 (3m - 1)}{(1 - m^2) T v_\phi} (1 + o(1)), \quad v_\phi = \frac{\sigma_e^2 + \phi^2 \sigma^2}{1 - m^2},$$

which is $O(1/T)$ and changes sign at $m = 1/3$.

Theorem 5 (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration policies with $\sum_{t \leq T} d_t^2 \leq D$ almost surely and all estimators (biased included), the minimax risk over an interval of width w satisfies*

$$\inf_{\text{policy, estimator}} \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\sigma^2}{D + \sigma^2(\pi/w)^2},$$

so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares.

Lemma 2 (Exploitation–information; L2). *For the symmetric two-point response family with half-width δ and any non-anticipating policy under the certainty-equivalent anchor,*

$$\mathbb{E}[\text{gain}_T] \leq \frac{g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4}}{\sigma^{1/2}}, \quad g_1 = \beta |h^* - \psi|,$$

via the exact identity $\mathbb{E}[\mathbb{E}[s \mid \mathcal{F}_t]] = \text{TV}(P_t^+, P_t^-)$ and Pinsker’s inequality $\text{TV}_t \leq \delta \sqrt{\mathbb{E} S_t} / \sigma$: exploitation is bounded by the information already purchased. At the minimax scale $\delta = c \sigma / \sqrt{\mathbb{E} S_T}$ this is $O(\sigma \sqrt{T})$, an $O(T^{-1/2})$ fraction of the $\Theta(T)$ exploration cost.

Theorem 6 (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator,*

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{\gamma_{\text{PO}} \sigma^2}{27} (1 - O(T^{-1/2})),$$

the supremum over the two-point response family at the minimax scale and the constant arising from the Le Cam reduction optimized at $c = 2/3$. The exploited rebate vanishes at rate $T^{-1/2}$ (Lemma L2). The sharp constant $\frac{1}{2}$, against which no simulated policy fell by more than Monte-Carlo error (`run_open1.py` §B), is conjectured and is part of OPEN-1.

Theorem 7 (Optimal exploration shapes; T5). *Under the value budget $\text{tr}(\Gamma_{\text{PO}} M) \leq B$ on the exploration second moment M : (a) the A-optimal design is $M^* = (B / \text{tr} \Gamma_{\text{PO}}^{1/2}) \Gamma_{\text{PO}}^{-1/2}$ with value $(\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$; (b) the D-optimal design is $M^* = (B/d) \Gamma_{\text{PO}}^{-1}$; (c) the c-optimal design for the correction functional c is $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$ (probing along the correction direction itself), with value $c^\top \Gamma_{\text{PO}} c / B$.*

Corollary 3 (Price of isotropic jitter; C5.1). *Isotropic exploration at matched budget overpays the A-optimal design by exactly the curvature-dispersion factor $F = d \text{tr}(\Gamma_{\text{PO}}) / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 \geq 1$, with equality iff all curvature directions are equal.*

Lemma 3 (Temporal shaping is irrelevant; L3). *With serially independent response noise, the information about the response parameters depends on the exploration process only through its empirical second moment; temporally correlated exploration buys nothing beyond its stationary design measure.*

Theorem 8 (Chebyshev unidentifiability; T6). *The sensitivity system of the structural family $\tau(h) = C_0 + C_1 e^{-ch}$ is an extended Chebyshev system; any design with fewer than three support points has singular Fisher information, and by C1.2 no retraining trajectory, at any modulus, identifies the response curvature c .*

Lemma 4 (Perturbed modulus; L4). *If the corrected update uses an estimated response with C^1 error e , the closed-loop modulus satisfies $|\hat{\rho} - \rho| \leq \eta \beta (\|e\|_\infty + \|h^* - \psi\| \|e'\|_\infty)$.*

Remark 2 (Open-loop neutrality; R2). *History-independent exploration schedules cannot change the linearized contraction rate of the retraining loop.*

Proposition 3 (Safety under pessimism; P7.1). *Given an anytime-valid confidence sequence for the response parameters at level $1 - \delta$, the pessimism rule (worst-case modulus over the confidence set kept below $1 - c_{\text{margin}}$, correction frozen otherwise) keeps the realized modulus within the margin for all t simultaneously with probability at least $1 - \delta$.*

Theorem 9 (Anchoring crossover, horizon-dependent; T7). *Under the value budget B and horizon T , the budget-and-horizon-optimal nonparametric (secant) estimator has*

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left(\frac{\tau'''(h^*)}{6} \right)^2 \left(\frac{2B}{\gamma_{\text{PO}} T} \right)^2,$$

so anchoring to a structural family with misspecification δ_{mis} wins iff (to leading order) $\delta_{\text{mis}} < |\tau'''(h^)| B / (3 \gamma_{\text{PO}} T)$: anchoring wins at short horizons and tight budgets, the free-form route asymptotically in T .*

Theorem 10 (The ROI of self-knowledge; T8). *Exploring once to accuracy v and correcting forever, discounted at ρ_{disc} , is optimal at $v^* = (\sigma/\kappa) \sqrt{\rho_{\text{disc}}}$, and exploring at all is worthwhile iff*

$$\rho_{\text{disc}} < \rho^* = \frac{(h_{\text{SP}} - h_{\text{PO}})^4}{4 \kappa^2 \sigma^2},$$

in which the curvature γ_{PO} cancels completely.

Proposition 4 (Frontier consistency; P9.1). *Every deterministic design sits exactly on the exchange-rate frontier; in particular the $O(\log T)$ -cost adaptive-design schedules of Lai–Robbins.*

Proposition 5 (Isotropy contrast; P9.2). *Naive isotropic exploration, optimal in isotropic LQR-like settings, overpays by exactly the dispersion factor F in the performative setting, with $F = 1$ iff the curvature spectrum is flat.*

Proposition 6 (Separable multi-bond curvature; P9.3). *In the separable multi-bond market, $(\Gamma_{\text{PO}})_{aa} = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a)$ exactly, with the factor-coupled case holding to first order in the off-diagonal response.*

B Proofs

Throughout, the local model of Assumptions A1–A6, per the derivation folder’s notation-and-assumptions register: **A1** (local quadratic scope): Φ is C^3 and strongly concave on the operating interval, expansions taken at h^* with third-order remainder bounded by $L_3 = \sup |\Phi'''|$; **A2** (exploration class): all policies are non-anticipating, with A2-sym additionally requiring $\mathbb{E}[u_t \mid \mathcal{F}_t] = 0$ and

324 A2-adaptive unrestricted, costs measured from the certainty-equivalent anchor; **A3** (noise exogeneity):
 325 ζ_t i.i.d., mean zero, variance σ^2 , independent of $\mathcal{F}_t$; **A4** (trust region): $|d_t| \leq r$ pathwise; **A5** (station-
 326 ary scale): $S_T = \Theta(T)$; **A6** (stable base loop): $m < 1$, equivalently $\rho(M) < 1$. The local dynamics
 327 are $d_{t+1} = -m d_t + u_t$ with $d_t = h_t - h^*$ (h^* the certainty-equivalent operating point, equal to
 328 h_{SP} for the blind loop), observations $\tau_t = \tau(h^*) - \varepsilon d_t + \zeta_t$, and Φ satisfying $\Phi'(h^*) = \delta_0 \neq 0$ and
 329 $-\Phi'' = \gamma_{\text{PO}}$ at the operating point.

330 A. Cost equivalence and its fluctuations

331 *Proof of Lemma L1 (cost equivalence).* Taylor's theorem with Lagrange remainder gives, for each t ,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2} \gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

332 with $L_3 = \sup |\Phi'''|$ on the operating interval. Summing over $t \leq T$ and taking expectations, it
 333 remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = -m d_t + \mathbb{E}[u_t |$
 334 $\mathcal{F}_t] = -m d_t$, so $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0] = 0$ when
 335 the loop starts at the operating point or in the stationary regime. The quadratic term contributes
 336 $\frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[\sum_t d_t^2]$, which is the claim.

337 For the variance decomposition, write the realized cost (net of the cubic remainder) as $C =$
 338 $-\delta_0 X + \frac{1}{2} \gamma_{\text{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\text{Var}(C) = \delta_0^2 \text{Var}(X) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(Y) -$
 339 $\delta_0 \gamma_{\text{PO}} \text{Cov}(X, Y)$. Under Gaussian innovations the vector $(d_1, \dots, d_T)$ is jointly Gaussian with
 340 mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\text{Cov}(X, Y) =$
 341 $\sum_{t,s} \mathbb{E}[d_t d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the
 342 same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3,
 343 including the necessity check that an asymmetric rule breaks the expectation identity by exactly
 344 $-\delta_0 \mathbb{E}[\sum_t d_t]$. $\square$

345 B. Saturation

346 *Proof of Theorem T1 (information saturation).* Let $\rho(M) < 1$ and define $P = \sum_{t \geq 0} (M^\top)^t M^t$.
 347 The series converges absolutely: by Gelfand's formula $\|M^t\| \leq C \rho_0^t$ for any $\rho(M) < \rho_0 < 1$ and
 348 some C , so the terms are dominated by $C^2 \rho_0^{2t}$. Termwise, $M^\top P M = \sum_{t \geq 0} (M^\top)^{t+1} M^{t+1} =$
 349 $P - I$, so P solves the discrete Lyapunov equation, and if Q is any other solution then $P - Q =$
 350 $M^\top (P - Q) M = (M^\top)^k (P - Q) M^k \rightarrow 0$, giving uniqueness. The energy identity is immediate:
 351 $\sum_{t \geq 0} \|M^t d_0\|^2 = d_0^\top (\sum_t (M^\top)^t M^t) d_0 = d_0^\top P d_0$. The directional version replaces I by $uu^\top$:
 352 $P_u = \sum_t (M^\top)^t uu^\top M^t$ solves $P_u = uu^\top + M^\top P_u M$ and $\sum_t (u^\top M^t d_0)^2 = d_0^\top P_u d_0$.

353 For the information claim: under A3 the log-likelihood of the observations along the noiseless
 354 trajectory has Fisher information for the response component along u equal to $\sigma^{-2} \sum_t (u^\top d_t)^2 =$
 355 $\sigma^{-2} d_0^\top P_u d_0$, which is finite and independent of the horizon; the Cramér–Rao bound gives the
 356 stated variance floor for unbiased estimators (the biased/minimax version is Theorem T3's van Trees
 357 argument). Verified: V1.1, V1.2, V1.4. $\square$

358 *Proof of Corollary C1.1 (safety implies blindness).* In the scalar case $P = \sum_t m^{2t} = (1 - m^2)^{-1}$,
 359 so the cap $(h_0 - h^*)^2 / (1 - m^2)$ has derivative in m equal to $2m (h_0 - h^*)^2 / (1 - m^2)^2 > 0$ for
 360 $m \in (0, 1)$: strictly increasing. $\square$

361 *Proof of Corollary C1.2 (design degeneracy).* For $m < 1$, $d_t = (-m)^t d_0$, so for every $\epsilon > 0$ all
 362 but $\lceil \log(|d_0|/\epsilon) / \log(1/m) \rceil$ of the points lie in $[-\epsilon, \epsilon]$: the empirical design measure converges
 363 weakly to δ_0 . At $m = 1$, $d_t = (-1)^t d_0$ and the design measure is $\frac{1}{2}(\delta_{+d_0} + \delta_{-d_0})$ exactly. In
 364 both cases the limiting design has at most two support points; combined with Theorem T6, the
 365 per-observation Fisher matrix of the three-parameter structural family is asymptotically singular
 366 along the trajectory. $\square$

367 *Proof of Remark R1 (non-normal transient information).* The identity $E(d_0) = d_0^\top P d_0$ of T1 holds
 368 for any stable M , normal or not. If $M = V \Lambda V^{-1}$ is diagonalizable, $\|M^t\| \leq \kappa(V) \rho(M)^t$

gives $\|P\| \leq \kappa(V)^2/(1 - \rho(M)^2)$. The bound is attained in order by the Jordan-type family
 $M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix}$ with $d_0 = e_2$: $M^t e_2 = (t a m^{t-1}, m^t)^\top$, so

$$E(e_2) = a^2 \sum_{t \geq 1} t^2 m^{2t-2} + \sum_{t \geq 0} m^{2t} = a^2 \frac{1 + m^2}{(1 - m^2)^3} + \frac{1}{1 - m^2},$$

using $\sum_{t \geq 1} t^2 x^{t-1} = (1 + x)/(1 - x)^3$ at $x = m^2$: the energy grows as a^2 , without bound, at fixed
spectral radius. Verified: V1.3 (the exact value matches the Lyapunov solve; 1268.2 vs the normal
cap 2.78 at $m = 0.8, a = 6$). $\square$

C. The exchange rate

Proof of Theorem T2 (pathwise exchange rate). Under A3, conditionally on the design $\{d_t\}$ the ob-
servations are independent with means linear in ε and variance σ^2 , so the least-squares slope has
conditional variance $\text{Var}(\hat{\varepsilon} \mid \text{design}) = \sigma^2/S_{xx}$ (Gauss–Markov). Multiplying by the quadratic cost
 $\frac{1}{2} \gamma_{\text{PO}} \sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T \bar{d}^2$ gives the displayed identity exactly, on every realization.
In the stationary regime $\bar{d} = O_P(T^{-1/2})$ by Proposition P2.1’s long-run variance, while $S_{xx}/T \rightarrow v$
a.s., so $T \bar{d}^2/S_{xx} = O_P(1/T)$. Verified: V2.1 (machine precision). $\square$

Proof of Proposition P2.1 (concentration). For the stationary AR(1) with lag-one coefficient $\rho =$
 $-m$, $\text{Cov}(d_t, d_s) = v(-m)^{|t-s|}$, so

$$\text{Var}\left(\sum_{t \leq T} d_t\right) = \sum_{t,s} v(-m)^{|t-s|} = Tv \frac{1+\rho}{1-\rho} (1 + o(1)) = Tv \frac{1-m}{1+m} (1 + o(1)),$$

the negative autocorrelation of the cobweb shrinking the fluctuation below the i.i.d. level. The realized-
cost product then fluctuates around the constant with standard deviation $O(T^{-1/2})$ by Lemma L1’s
variance decomposition (both $\text{Var}(\sum d)$ and $\text{Var}(\sum d^2)$ are $O(T)$ while the squared mean of the
cost is $O(T^2)$). Verified: V2.2 (variance ratio 4.80 for a $4 \times$ horizon), V2.3. $\square$

Proof of Proposition P2.2 (feedback bias). Under A3’, $d_{t+1} = -m d_t + \sigma_e \xi_t + \phi \zeta_t$, so d_t is a
linear function of past innovations with $\mathbb{E}[d_s \zeta_t] = \phi \sigma^2 (-m)^{s-t-1}$ for $s > t$ and 0 otherwise.
Write the slope error as N/S with $N = \sum_t (d_t - \bar{d}) \zeta_t$ and $S = S_{xx}$, and expand $\mathbb{E}[N/S] =$
 $\mathbb{E}[N]/\mathbb{E}[S] - \text{Cov}(N, S)/\mathbb{E}[S]^2 + O(T^{-2})$ (valid for stationary Gaussian data; the neglected term
involves $\mathbb{E}[N] \text{Var}(S)/\mathbb{E}[S]^3 = O(T^{-2})$).

Centering term. $\mathbb{E}[d_t \zeta_t] = 0$, so $\mathbb{E}[N] = -\frac{1}{T} \sum_{s>t} \mathbb{E}[d_s \zeta_t] = -\frac{1}{T} \sum_{s>t} \phi \sigma^2 (-m)^{s-t-1} =$
 $-\phi \sigma^2/(1+m) + O(1/T)$.

Covariance term. For jointly Gaussian zero-mean (X, Y, Z) , $\mathbb{E}[XY^2Z] = 2\mathbb{E}[XY]\mathbb{E}[YZ] +$
 $\mathbb{E}[XZ]\mathbb{E}[Y^2]$ (Isserlis). Applying this to $\text{Cov}(N, S) = \sum_{t,s} (\mathbb{E}[\tilde{d}_t \tilde{d}_s^2 \zeta_t] - \mathbb{E}[\tilde{d}_t \zeta_t] \mathbb{E}[\tilde{d}_s^2])$ with
 $\tilde{d} = d - \bar{d}$ leaves only the paired term $2 \sum_{t,s} \mathbb{E}[\tilde{d}_t \tilde{d}_s] \mathbb{E}[\tilde{d}_s^2 \zeta_t]$; substituting the stationary covariances,

$$\text{Cov}(N, S) = 2T v_\phi \phi \sigma^2 \sum_{k \geq 1} (-m)^k (-m)^{k-1} + O(1) = -\frac{2T v_\phi \phi \sigma^2 m}{1 - m^2} + O(1).$$

Combining with $\mathbb{E}[S] = Tv_\phi (1 + o(1))$:

$$\mathbb{E}[\hat{\varepsilon} - \varepsilon_{\text{true}}] = \frac{\phi \sigma^2}{Tv_\phi} \left[-\frac{1}{1+m} + \frac{2m}{1-m^2} \right] + O(T^{-2}) = \frac{\phi \sigma^2 (3m-1)}{(1-m^2)Tv_\phi} + O(T^{-2}).$$

The bracket changes sign at $m = 1/3$. (The first draft kept only the centering term and was falsified
by the verification suite; the record is in VERIFICATION.md.) Verified: V2.4, V2.5 – both signs,
straddling $m = 1/3$. $\square$

D. Minimax lower bounds

Proof of Theorem T3 (deviation-budget minimax). Let π be a prior density on an interval of width w with finite information $I(\pi) = \int (\pi')^2 / \pi$; the cosine-squared density attains the minimum $I(\pi) = (\pi/w)^2$ (here the first π is the constant). For any non-anticipating policy the joint density of the data factorizes as $\prod_t p(u_t | \mathcal{F}_t) p(\tau_t | \mathcal{F}_t, \varepsilon)$, where the policy factors do not depend on ε and $\tau_t | \mathcal{F}_t \sim \mathcal{N}(\tau(h^*) - \varepsilon d_t, \sigma^2)$ with d_t being $\mathcal{F}_t$ -measurable. Hence the Fisher information of the whole experiment is

$$I_T(\varepsilon) = \mathbb{E}_\varepsilon \left[\sum_t (\partial_\varepsilon \log p(\tau_t | \mathcal{F}_t, \varepsilon))^2 \right] = \mathbb{E}_\varepsilon \left[\sum_t d_t^2 \right] / \sigma^2 \leq D / \sigma^2$$

under the pathwise budget: adaptivity places information but cannot manufacture it. The van Trees inequality (Gill–Levit) then bounds the Bayes risk of *any* estimator, biased or not: $\mathbb{E}_\pi \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq (\mathbb{E}_\pi[I_T] + I(\pi))^{-1} \geq (D/\sigma^2 + (\pi/w)^2)^{-1}$, and the supremum over the interval dominates the prior average. Equality (up to the prior term) is attained by symmetric two-point probing at the budget with least squares, by Theorem T2. Verified: V3.1, V3.2 (equality within Monte-Carlo error). $\square$

Proof of Lemma L2 (exploitation–information). Prior $s = \pm 1$ symmetric; environments $P^\pm$ differing only in the observation means by $2\delta d_t$ at step t . Three ingredients.

(i) *Posterior imbalance equals total variation in expectation.* Under the symmetric mixture $\bar{P} = \frac{1}{2}(P^+ + P^-)$, with data x ,

$$\mathbb{E}_{\bar{P}} |\mathbb{P}(s=+ | x) - \mathbb{P}(s=- | x)| = \int \frac{1}{2}(dP^+ + dP^-) \frac{|dP^+ - dP^-|}{dP^+ + dP^-} = \text{TV}(P_t^+, P_t^-).$$

(ii) *Pinsker with the chain rule for adaptive designs.* $\text{KL}(P_t^+ \| P_t^-) = \mathbb{E}^+ [\sum_{i \leq t} (2\delta d_i)^2 / (2\sigma^2)] = 2\delta^2 \mathbb{E}^+[S_t] / \sigma^2$ (the policy factors cancel inside the log-ratio), so $\text{TV}_t \leq \delta \sqrt{\mathbb{E}[S_t]} / \sigma$.

(iii) *The gain bound.* Since d_t is $\mathcal{F}_t$ -measurable, $\mathbb{E}[s d_t] = \mathbb{E}[d_t \mathbb{E}[s | \mathcal{F}_t]]$; two applications of Cauchy–Schwarz and $|\mathbb{E}[s | \mathcal{F}_t]| \leq 1$ give

$$\mathbb{E}[\text{gain}_T] = g_1 \delta \sum_t \mathbb{E}[s d_t] \leq g_1 \delta \sum_t (\mathbb{E}d_t^2)^{1/2} \text{TV}_t^{1/2} \leq g_1 \delta (\mathbb{E}S_T)^{1/2} (T \text{TV}_T)^{1/2},$$

and inserting (ii): $\mathbb{E}[\text{gain}_T] \leq g_1 \delta^{3/2} T^{1/2} (\mathbb{E}S_T)^{3/4} / \sigma^{1/2}$.

Remark on constants. The derivation document’s heuristic chain (treating the imbalance as bounded by TV_t pathwise) yields $g_1 \delta^2 S_T \sqrt{T} / \sigma$; the rigorous chain above changes the exponent bookkeeping but not the conclusion of Theorem T4: at the minimax scale $\delta = c \sigma / \sqrt{\mathbb{E}S_T}$ both give $\mathbb{E}[\text{gain}_T] = O(\sigma \sqrt{T})$, hence an exploited fraction $O(T^{-1/2})$ of the $\Theta(T)$ cost under A5. Verified: V3.3 (the imbalance–TV inequality pathwise in expectation), V3.4 (the $T^{-1/2}$ decay at the minimax scale). $\square$

Proof of Theorem T4 (value-budget minimax; constant-factor version). Fix a policy and estimator; let $\bar{S} = \mathbb{E}[S_T] = \Theta(T)$ under A5. Choose the two-point prior at separation $\delta = c \sigma / \sqrt{\bar{S}}$ with $c \in (0, 1)$ to be optimized. By the testing reduction (Le Cam), any estimator’s worst-case risk over the pair obeys $\sup_\pm \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\delta^2}{2} (1 - \text{TV}_T) \geq \frac{\delta^2}{2} (1 - c)$ using Lemma L2(ii). By Lemma L1 with the certainty-equivalent anchor and Lemma L2’s gain bound, the value cost obeys $C_T \geq \frac{1}{2} \gamma_{\text{PO}} \bar{S} (1 - O(T^{-1/2}))$. Multiplying,

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{c^2 \sigma^2}{2\bar{S}} (1 - c) \cdot \frac{\gamma_{\text{PO}} \bar{S}}{2} (1 - O(T^{-1/2})) = \frac{\gamma_{\text{PO}} \sigma^2}{4} c^2 (1 - c) (1 - O(T^{-1/2})),$$

maximized at $c = 2/3$: the bound $\gamma_{\text{PO}} \sigma^2 / 27 \cdot (1 - O(T^{-1/2}))$.

Status. This proves the value-budget product is bounded below by an explicit constant multiple of the exchange rate, uniformly over adaptive policies, with the exploited rebate vanishing at rate $T^{-1/2}$. The sharp constant $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$ – numerically supported (run_open1.py §B: three adaptive amplitude schedules, ratios 0.995 to 1.018) – requires the van Trees route with cost control in place of the two-point reduction and is part of OPEN-1. $\square$

439 E. Design geometry

440 *Proof of Theorem T5 (optimal exploration shapes).* All three problems have convex (resp. concave)
441 objectives over the PSD cone with one linear constraint; KKT stationarity plus convexity/concavity
442 certifies global optimality.

443 (a) *A-optimality.* $\nabla_M \text{tr}(M^{-1}) = -M^{-2}$, so stationarity gives $M^{-2} = \lambda \Gamma_{\text{PO}}$, i.e. $M =$
444 $\lambda^{-1/2} \Gamma_{\text{PO}}^{-1/2}$; the budget $\text{tr}(\Gamma_{\text{PO}} M) = \lambda^{-1/2} \text{tr}(\Gamma_{\text{PO}}^{1/2}) = B$ pins $\lambda^{-1/2} = B / \text{tr}(\Gamma_{\text{PO}}^{1/2})$
445 and the value $\text{tr}(M^{-1}) = (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$. (Diagonal-case cross-check by Cauchy–Schwarz:
446 $(\sum_i 1/m_i)(\sum_i g_i m_i) \geq (\sum_i \sqrt{g_i})^2$ with equality at $m_i \propto g_i^{-1/2}$.)

447 (b) *D-optimality.* $\nabla_M \log \det M = M^{-1} = \lambda \Gamma_{\text{PO}}$ gives $M = (B/d) \Gamma_{\text{PO}}^{-1}$ after normalizing by the
448 budget.

449 (c) *c-optimality.* Substitute $N = \Gamma_{\text{PO}}^{1/2} M \Gamma_{\text{PO}}^{1/2}$ ($\text{tr} N = \text{tr}(\Gamma_{\text{PO}} M) \leq B$) and $w = \Gamma_{\text{PO}}^{1/2} c$,
450 so the objective is $w^\top N^{-1} w$. By Cauchy–Schwarz, $(w^\top w)^2 = (w^\top N^{-1/2} N^{1/2} w)^2 \leq$
451 $(w^\top N^{-1} w)(w^\top N w)$ and $w^\top N w \leq \|w\|^2 \text{tr} N \leq \|w\|^2 B$, hence $w^\top N^{-1} w \geq \|w\|^2 / B =$
452 $c^\top \Gamma_{\text{PO}} c / B$, attained at $N^* = B w w^\top / \|w\|^2$. Back-substituting with $\Gamma_{\text{PO}}^{-1/2} w = c$ gives
453 $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$; the optimal probe direction is c itself, the curvature pricing only its
454 scale. Verified: V4.1–V4.3 (formulas achieved; never beaten by 4000 random feasible designs
455 each). $\square$

456 *Proof of Corollary C5.1 (price of isotropic jitter).* Isotropic $M = (B / \text{tr} \Gamma_{\text{PO}}) I$ has $\text{tr}(M^{-1}) =$
457 $d \text{tr}(\Gamma_{\text{PO}}) / B$; dividing by the optimum $(\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$ gives $F = d \text{tr}(\Gamma_{\text{PO}}) / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2$, and $F \geq 1$
458 with equality iff all eigenvalues are equal, by Cauchy–Schwarz applied to $(\sum_i \sqrt{g_i} \cdot 1)^2 \leq d \sum_i g_i$.
459 Verified: V4.4. $\square$

460 *Proof of Lemma L3 (temporal shaping).* Under A3 the conditional log-likelihood of the responses
461 given the deployment path is $-\sum_t (\tau_t - a - \theta^\top d_t)^2 / (2\sigma^2)$ up to constants; its Hessian in θ is
462 $-\sum_t d_t d_t^\top / \sigma^2$, a function of the empirical second moment only, whatever the temporal dependence
463 of $\{d_t\}$. Verified: V4.5. $\square$

464 *Proof of Theorem T6 (Chebyshev unidentifiability).* The sensitivity of the family $\tau(h) = C_0 +$
465 $C_1 e^{-ch}$ is $s(h) = (1, e^{-ch}, -C_1 h e^{-ch})^\top$. The functions $\{1, e^{-ch}, h e^{-ch}\}$ form an extended
466 Chebyshev system on any interval (their Wronskian is $c^3 e^{-2ch} \neq 0$ after row reduction; equivalently,
467 a nontrivial linear combination $a + (b + \gamma h) e^{-ch}$ has at most two zeros counting multiplicity). For
468 any design ξ with k support points, the Fisher matrix $M(\xi) = \sum_{i \leq k} w_i s(h_i) s(h_i)^\top$ has rank at
469 most k ; with $k \leq 2 < 3$ it is singular, and that null space always loads on the c -sensitivity: writing n
470 for the null vector of the two-point design matrix, $n_3 = 0$ would force $n_1 + n_2 e^{-ch_i} = 0$ at both
471 support points, hence $n = 0$ for $h_1 \neq h_2$, a contradiction. By Corollary C1.2 the trajectory design
472 has at most two effective support points, which proves the claim; the existence of exactly-three-point
473 D-optimal designs on a trust-region interval is classical for Chebyshev systems (Karlin–Studden).
474 Verified: V5.1 (two-point determinants at 10^{-17} ; three-point at 6×10^{-3}). $\square$

475 F. Safety under certainty equivalence

476 *Proof of Lemma L4 (perturbed modulus).* The corrected map is $\Psi(h) = h + \eta [G(h) - \beta(h -$
477 $\psi) \hat{\varepsilon}(h)]$ and the ideal map replaces $\hat{\varepsilon}$ by ε . With $e = \hat{\varepsilon} - \varepsilon$, $\Psi'(h) - \Psi'_{\text{ideal}}(h) = -\eta \beta \frac{d}{dh} [(h -$
478 $\psi) e(h)] = -\eta \beta [e(h) + (h - \psi) e'(h)]$, and taking suprema over the operating interval gives
479 $|\hat{\rho} - \rho| \leq \eta \beta (\|e\|_\infty + |h - \psi|_{\max} \|e'\|_\infty)$. Verified: V6.1 (nine-cell error grid; the bound holds with
480 maximal excess 2.4×10^{-11}). $\square$

481 *Proof of Remark R2 (open-loop neutrality).* For $d_{t+1} = -m d_t + u_t$ with a history-independent
482 schedule $\{u_t\}$, two solutions with different initial conditions differ by the homogeneous solution
483 $(-m)^t (d_0 - d'_0)$, which contracts at rate m regardless of the schedule (superposition). Verified: V6.2
484 (difference exactly 0 after 200 steps to machine precision). $\square$

485 *Proof of Proposition P7.1 (safety under pessimism).* Let E be the event that the confidence sequence
486 contains the true response parameters simultaneously for all t ; by anytime-validity $\mathbb{P}(E) \geq 1 - \delta$.
487 On E , the true parameters lie in Θ_t at every step, so the realized modulus is dominated by the rule's
488 worst case: $|\rho_t| \leq \sup_{\theta \in \Theta_t} |\hat{\rho}(\theta)| \leq 1 - c_{\text{margin}}$ whenever correction is active, and when the rule
489 freezes, the blind loop's modulus $m < 1 - c_{\text{margin}}$ (choose the margin below $1 - m$) applies. No
490 union bound over time is needed: the inclusion is pathwise on E . (*Conditional result: the hypothesis*
491 *is the confidence sequence's validity; constructing it from the D2 information accounting is routine,*
492 *and the necessity of the hypothesis is demonstrated numerically – an invalid, too-narrow sequence*
493 *produces violations.*) $\square$

494 G. The anchoring crossover

495 *Proof of Theorem T7 (horizon-dependent crossover).* *Bias.* For the symmetric secant with half-width
496 w , fifth-order Taylor expansion of $\tau(h^* \pm w)$ gives

$$\frac{\tau(h^* + w) - \tau(h^* - w)}{2w} = \tau'(h^*) + \frac{\tau'''(h^*)}{6} w^2 + O(w^4),$$

497 all even-order terms cancelling (symmetric probing is bias-optimal at its order; the second derivative
498 never enters). Verified: V7.1.

499 *Variance.* With n probe pairs ($T = 2n$ observations) the slope estimator $(\bar{y}_+ - \bar{y}_-)/(2w)$ has variance
500 $\frac{2\sigma^2/n}{4w^2} = \frac{\sigma^2}{2nw^2} = \frac{\sigma^2}{S}$ with $S = \sum_t d_t^2 = 2nw^2 = Tw^2$: variance depends on the design only
501 through its energy.

502 *Constrained optimum.* The value budget fixes $S = 2B/\gamma_{\text{PO}}$; the horizon caps the number of
503 observations, $Tw^2 \geq S$, i.e. $w^2 \geq 2B/(\gamma_{\text{PO}}T)$; bias is increasing in w , so the constrained optimum
504 sits at $w^2 = 2B/(\gamma_{\text{PO}}T)$, giving

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left(\frac{\tau'''(h^*)}{6} \right)^2 \left(\frac{2B}{\gamma_{\text{PO}}T} \right)^2.$$

505 Verified: V7.2–V7.4 (three (B, T) cells, within 2%).

506 *Crossover.* The anchored estimator has $\text{MSE} = \kappa_p \gamma_{\text{PO}} \sigma^2 / (2B) + \delta_{\text{mis}}^2$; comparing and keeping the
507 dominant term gives the rule $\delta_{\text{mis}} < |\tau'''(h^*)| B / (3\gamma_{\text{PO}}T)$ to leading order, with the exact two-term
508 form in the display above. The direction of the comparison on both sides of the threshold is verified:
509 V7.5, V7.6. $\square$

510 H. ROI and instantiation

511 *Proof of Theorem T8 (ROI of self-knowledge).* With $A = \frac{1}{2}\gamma_{\text{PO}}(h_{\text{SP}} - h_{\text{PO}})^2$, $a = \frac{1}{2}\gamma_{\text{PO}}\kappa^2$, $b =$
512 $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$, the net present value of exploring to accuracy v is $\text{NPV}(v) = (A - av)/\rho_{\text{disc}} - b/v$, strictly
513 concave on $(0, \infty)$ with $\text{NPV}'(v) = -a/\rho_{\text{disc}} + b/v^2$, giving $v^* = \sqrt{b\rho_{\text{disc}}/a} = (\sigma/\kappa)\sqrt{\rho_{\text{disc}}}$
514 (the curvature cancels between a and b). Substituting, $\text{NPV}(v^*) = A/\rho_{\text{disc}} - 2\sqrt{ab/\rho_{\text{disc}}}$, which
515 is positive iff $A > 2\sqrt{ab\rho_{\text{disc}}}$, i.e. iff $\rho_{\text{disc}} < A^2/(4ab) = (h_{\text{SP}} - h_{\text{PO}})^4/(4\kappa^2\sigma^2)$: γ_{PO} cancels
516 completely. Verified: V8.1, V8.2 (the break-even root to 10^{-3} relative). $\square$

517 *Proof of Proposition P9.1 (frontier consistency).* For any deterministic design, $\text{Var}(\hat{\varepsilon} \mid \text{design}) \cdot$
518 $\frac{1}{2}\gamma_{\text{PO}}S = (\sigma^2/S) \cdot \frac{1}{2}\gamma_{\text{PO}}S = \frac{1}{2}\gamma_{\text{PO}}\sigma^2$: the design energy cancels identically, whatever the schedule
519 – including the $d_t \propto t^{-1/2}$ schedules with $S_T = \Theta(\log T)$ of the Lai–Robbins adaptive-design line.
520 Verified: V8.3 (exact). $\square$

521 *Proof of Proposition P9.2 (isotropy contrast).* Immediate from Corollary C5.1: $F = 1$ iff the cur-
522 vature spectrum is flat – the isotropic (LQR-like) case in which naive exploration is optimal; any
523 curvature dispersion makes $F > 1$ by the strict case of Cauchy–Schwarz. Cross-verified with
524 V4.4. $\square$

525 *Proof of Proposition P9.3 (separable multi-bond curvature).* In the separable case the objective is a
526 sum of per-bond objectives $\Phi_a(h_a)$, each of the scalar form analyzed in the base project's theory

1.2, so the Hessian is diagonal and the scalar computation applies per bond: differentiating $\Phi_a(h) = P[hA_a e^{-k_a h} \rho + h\tau_a(h) - \psi_a \tau_a(h) - w_a(h - h_{\text{ref}})^2]$ twice and collecting terms with $\tau'_a = -\varepsilon_a$, $\tau''_a = c_t \varepsilon_a$ yields $-\Phi''_a = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a)$, which is the claimed diagonal entry. In the factor-coupled case the off-diagonal response contributes additional terms of first order in $\|E_{\text{offdiag}}\|$, stated as a construction with its error term (not claimed exact). Verified: V8.4 (finite-difference Hessians of three random bonds, relative error below 10^{-4}). $\square$

C Selected verification rows

Table 1: Selected verification rows (full table, seeds, and code in the repository).

Claim (register)	Measured	Predicted	
Exchange rate, $m=0.3$, two amplitudes (T2)	0.01062, 0.01070	0.01056	pass
Exchange rate, $m=0.6$, two amplitudes (T2)	0.00549, 0.00569	0.00535	pass
Saturating environment (outside A1)	+16% over rate	—	drift
Feedback floor, $m=0.3/0.6/0.85$ (T1')	within 1–7%	$(\sigma/\gamma)^2/(1-m^2)$	pass
Design arm $\text{sd}(\hat{c})$; jitter/design ratio (T6)	0.234; $7.8\times$	Fisher 0.248; ≥ 3	pass
SafeD reaches h_{PO} ; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, $d=8$ (T5)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Pricing domain, 4 cells (T2, exact: LQ); agent	0.384, 0.192; 12.65	0.384, 0.192; 12.5	pass
Multi-bond shaped/iso steady-state risk (T5)	1.20 (flat control 1.00)	$> 1.15 (\approx 1)$	pass
Structure-proof floor, min CR ratio (OPEN-1)	1.005	≥ 1	pass

D Additional figures

E Reproducibility

All results are deterministic from seeds, numpy-only, CPU-only; reported final errors are 300-step tail averages over the deployed path, stable to three decimals under horizon perturbation. The repository contains: the derivation documents and theorem register (25 results; 34 numerical derivation checks), the pipeline (environments, estimators, designs, agents, baselines; 36 measured-vs-predicted rows; 9 unit tests), the real-data leg with its 10/10 port validation, and continuous integration running the verification suites on every push. The register of ten measurement-forced pivots, each recorded in the code where it happened, with the failed prediction and the fix, is in the repository README.

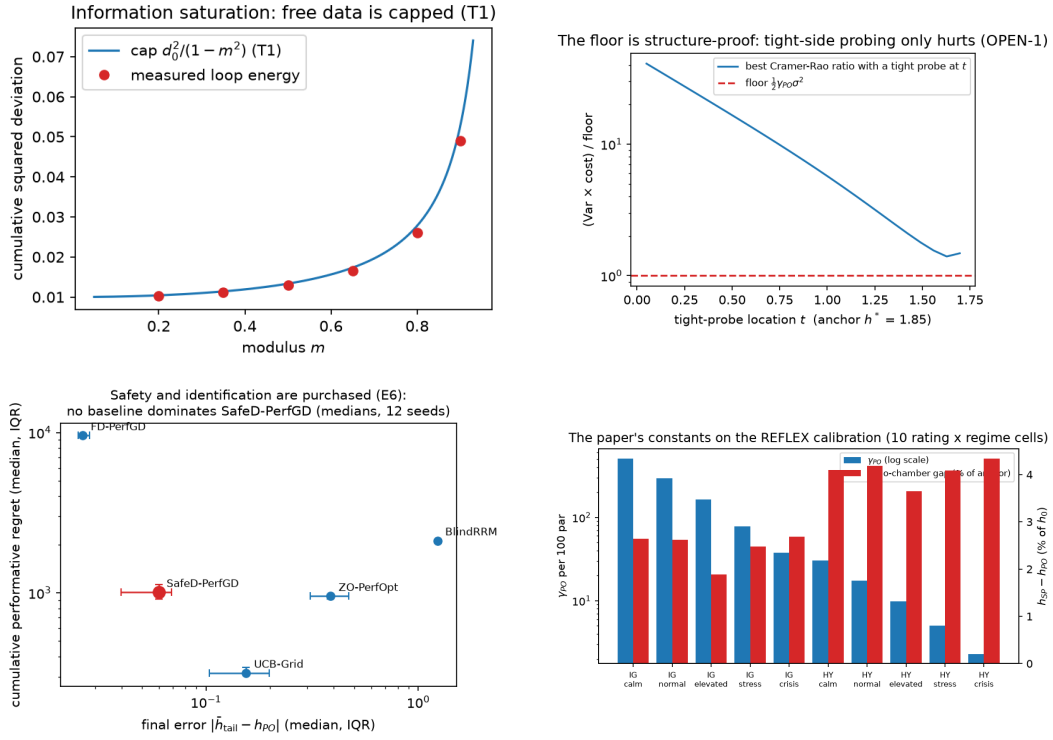


Figure 2: **Top left:** information saturation: measured loop energy against the cap $d_0^2/(1-m^2)$ (T1). **Top right:** the floor is structure-proof: best Cramér–Rao ratio with a tight probe at t ; tight-side probing only hurts (OPEN-1). **Bottom left:** baseline Pareto frontier: no baseline dominates SafeD-PerfGD on (final error, regret). **Bottom right:** the paper’s constants on the real-data calibration, 10 rating $\times$ regime cells.